

Problem-Solution Fit

Project Information

Date	15 March 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas

1. Customer Segments	2. Problems / Pains
Primary users include farmers, agricultural students, and small-scale cultivators. Secondary users include agricultural researchers, agribusiness companies, universities, government agencies, and policymakers.	• Farmers struggle to select the most suitable crop for their land. • Soil nutrients and climatic conditions vary by region. • Incorrect crop selection causes low yield, water wastage, fertilizer misuse, and financial loss.
3. Triggers to Act	4. Emotions Before / After
Start of cultivation season, poor previous harvest, changing rainfall patterns, fertilizer planning, and the need for profitable crop selection.	Before: Uncertain, worried, dependent on guesswork, and concerned about crop failure. After: Confident, informed, prepared, and supported by data-driven recommendations.
5. Available Solutions	6. Customer Limitations
Traditional farming experience, local expert advice, agriculture department guidance, and general online resources.	Limited access to experts, lack of real-time soil analysis, unstable internet in rural areas, limited technical knowledge, and budget constraints.
7. Behavior	8. Channels

Farmers usually depend on past experience, neighboring farmers, seasonal patterns, and local market demand before choosing crops.	Online: Web application, GitHub repository, agriculture portals, and government advisory platforms. Offline: Field observations, local records, soil reports, and consultation with agricultural officers.
9. Root Cause	10. Your Solution
Crop selection is often based on trial-and-error methods rather than scientific analysis of soil nutrients and environmental conditions.	OptiCrop: Smart Agricultural Production Optimization Engine uses Machine Learning to analyze Nitrogen, Phosphorus, Potassium, temperature, humidity, pH, and rainfall values. It recommends the most suitable crop with a confidence score, crop information, growing season, water requirement, and fertilizer suggestion.